

report

```
-> checkpoint1 - multiple fork: 6, pid: 40, free pages: 607
-> checkpoint1 - multiple fork: 7, pid: 41, free pages: 597
-> checkpoint1 - multiple fork: 8, pid: 42, free pages: 587
-> checkpoint1 - multiple fork: 9, pid: 43, free pages: 577
-> checkpoint1 - multiple fork: 0, pid: 34, exited
-> checkpoint1 - multiple fork: 1, pid: 35, exited
-> checkpoint1 - multiple fork: 2, pid: 36, exited
-> checkpoint1 - multiple fork: 3, pid: 37, exited
-> checkpoint1 - multiple fork: 4, pid: 38, exited
-> checkpoint1 - multiple fork: 5, pid: 39, exited
-> checkpoint1 - multiple fork: 6, pid: 40, exited
-> checkpoint1 - multiple fork: 7, pid: 41, exited
-> checkpoint1 - multiple fork: 8, pid: 42, exited
-> checkpoint1 - multiple fork: 9, pid: 43, exited
-> checkpoint1 - passed
sh > child 3 exit with code 0
sh >>
```

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-> checkpoint2 - multiple fork: 0, pid: 75, exited
-> checkpoint2 - multiple fork: 1, pid: 76, exited
-> checkpoint2 - multiple fork: 2, pid: 77, exited
-> checkpoint2 - multiple fork: 3, pid: 78, exited
-> checkpoint2 - multiple fork: 4, pid: 79, exited
-> checkpoint2 - multiple fork: 5, pid: 80, exited
-> checkpoint2 - multiple fork: 6, pid: 81, exited
-> checkpoint2 - multiple fork: 7, pid: 82, exited
-> checkpoint2 - multiple fork: 8, pid: 83, exited
-> checkpoint2 - multiple fork: 9, pid: 84, exited
-> checkpoint2 - passed
-> checkpoint2 - check whether you leak any page
-> checkpoint2 - passed
sh > child 44 exit with code 0
sh >>
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[INFO 0,-1] load_init_app: load init proc init
[INFO 0,-1] bootcpu_init: start scheduler!
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init: starting sh

sh >> cowtest 3

```
-> checkpoint3 - allocate heap: 50 pages, filling with mypid
-> checkpoint3 - CoW fork, child read
-> checkpoint3 - CoW fork, child write
-> checkpoint3 - CoW fork, child and parent write
-> checkpoint3 - check whether you leak any page
-> checkpoint3 - passed
sh > child 3 exit with code 0
sh >> █
```

```
setup: remaining free pages: 473, initial free pages: 29374
setup: remaining free pages: 373, initial free pages: 29374
setup: remaining free pages: 273, initial free pages: 29374
setup: remaining free pages: 173, initial free pages: 29374
-> checkpoint4 - allocate heap: 50 pages, filling with mypid
-> checkpoint4 - CoW fork, copy_to_user should also do CoW
-> checkpoint4 - CoW fork, read-only remains read-only
[INFO 0,303] handle_pgfault: page fault in application, bad addr = 0x000000000000
4056a8, bad instruction = 0x000000000000403000, core dumped.
-> checkpoint4 - check whether you leak any page
-> checkpoint4 - passed
sh > child 8 exit with code 0
sh >> 
```

用时:4小时